

Literature status: direct-product stability of the M_d -approximation property

Run fa_banach_001, lane 2

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Status. `literature_already_answered`. This packet records an exact later answer and does not claim a new mathematical result.

Source question. In Remark 4.2 on PDF page 7 of Vergara’s 2023 paper [1], the author writes: “We do not know if M_d -AP is stable under direct products.” The setting is discrete groups, with $M_d(G) = X_d(G)^*$ and M_d -AP meaning that the constant function 1 lies in the $\sigma(M_d(G), X_d(G))$ -closure of $\mathbb{C}[G]$.

Exact later answer. Vergara’s 2025 follow-up [2] explicitly cites the source and proves a stronger extension theorem. Its Theorem 1.3 states that, for a normal subgroup $\Gamma \triangleleft G$ and $d \geq 2$, if both Γ and G/Γ have M_d -AP, then G has M_d -AP. Corollary 1.4 on PDF page 3 states explicitly that M_d -AP is stable under direct products, semidirect products, and free products. The proof on PDF page 10 obtains the direct-product case from the semidirect-product case with trivial action. Thus Remark 4.2 is answered affirmatively.

Independent consistency check. If $u_i \rightarrow 1$ weak-* in $M_d(G)$ and $v_j \rightarrow 1$ weak-* in $M_d(H)$ are finitely supported, set $w_{i,j}(g, h) = u_i(g)v_j(h)$. Tensoring the defining Hilbert-space factorizations gives

$$\|w_{i,j}\|_{M_d(G \times H)} \leq \|u_i\|_{M_d(G)} \|v_j\|_{M_d(H)}.$$

Both approximating nets are norm bounded. The product net converges pointwise to 1 and therefore converges on $\mathbb{C}[G \times H]$; its uniform norm bound and the density of this subspace in $X_d(G \times H)$ upgrade the convergence to weak-* convergence. This verifies the direct-product statement from the definitions, but it is not a novelty claim.

Scope. Only the direct-product question in Remark 4.2 is settled here. The broader question whether M_d -AP is equivalent to ordinary AP remains open; the 2025 paper restates it as the equivalence of M_2 -AP and M_d -AP for all $d \geq 3$.

References

- [1] I. Vergara, *The M_d -approximation property and unitarisability*, Proc. Amer. Math. Soc. 151 (2023), 1209–1220; [arXiv:2202.12198](#).
- [2] I. Vergara, *Some remarks on M_d -multipliers and approximation properties*, [arXiv:2509.07861](#) (2025).